CogniNet-6G: An AI-Native Closed-Loop Digital Twin Framework for Predictive Semantic Network Orchestration in Private 6G Industrial Robot Networks

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Abstract High-bandwidth inspection feeds and reliable low-latency performance are essential as industrial mobile robots navigate automated shop floors. Sub-millisecond control for functional safety is equally vital. In high-frequency 28 GHz mmWave private cellular links, physical obstacles—such as moving steel containers, overhead gantry cranes, and tall storage racks—frequently induce line-of-sight discontinuities, producing rapid drops in received signal strength. Conventional network slicing schedulers reactively adjust Physical Resource Block (PRB) quotas based on link quality only after queues accumulate and packets drop. In a mobile robot fleet traveling at 1.5 to 2.0 m/s, this reaction lag frequently breaches communication keep-alive thresholds, triggering emergency mechanical braking that stalls production lines. In this work, we present CogniNet-6G: a closed-loop digital twin framework designed to forecast wireless link degradation and ensure safety-critical traffic remains protected before delays manifest. A 2-layer bidirectional Long Short-Term Memory (Bi-LSTM) network predicts slice demand across a 2.0 s lookahead horizon, paired with a sub-millisecond Multi-Layer Perceptron (MLP) packet classifier. An edge-deployed Proximal Policy Optimization (PPO) agent dynamically reassigns radio slices, while an analytical convex safety projection operator (Π_C) guarantees that the bandwidth allocation ($a_{\text{URLLC}} \geq 0.20$) remains reserved whenever safety-critical packets are in transit. We validated CogniNet-6G through co-simulations coupling ROS 2 Gazebo robot kinematics with discrete-event ns-3 28 GHz 5G-LENA mmWave channels across 10 operational factory stress scenarios (150 independent trials, 9,000 control steps). Experimental results demonstrate that CogniNet-6G dispatches emergency stop messages in 0.45 ± 0.95 ms, eliminates safety SLA violations under severe interference and structural shadowing, cuts reallocation thrashing by 60.2%, and runs in 2.82 ms on commodity edge hardware, comfortably within the 10 ms O-RAN Near-RT RIC scheduling envelope.

Keywords 6G Networks · Digital Twin · Reinforcement Learning · Semantic Communication · Network Slicing · Industrial Robots (AGV/AMR) · AI-Native Orchestration · Private Networks · O-RAN · Predictive QoS

1 Introduction

1.1 General Description

Modern manufacturing facilities are steadily replacing static mechanical conveyor systems with autonomous mobile robots (AMRs) and automated guided vehicles (AGVs) [13, 31]. In automotive assembly, battery gigafactories, and cleanrooms, mobile robot fleets transport components, inspect assemblies with high-definition vision cameras, and work alongside human technicians without physical protective fences [12, 21].

Because physical mobility prevents the use of wired connections, every movement trajectory, sensor update, cooperative maneuver, and emergency stop command must travel over a wireless medium [9, 35]. Private cellular networks deployed in millimetre-wave (mmWave) bands—specifically 28 GHz and sub-terahertz frequencies—have become the preferred wireless backbone for smart manufacturing [3, 4]. These deployments deliver gigabit-per-second uplink throughput with nominal air-interface delays under a millisecond, satisfying the nominal requirements of Ultra-Reliable Low-Latency Communication (URLLC) [9].

However, dense robot fleet operations over mmWave wireless links reveal serious protocol- and physical-level vulnerabilities [6, 11]. Directional propagation at 28 GHz relies on narrow beams and experiences high path loss [4]. Moving metallic objects, such as automated gantry cranes, forklift assemblies, and steel inventory racking, frequently obstruct line-of-sight (LOS) paths, resulting in abrupt 20 to 30 dB reductions in received signal strength. Simultaneously, factory processes produce bursty, synchronized data surges. When multiple AMRs meet at a narrow aisle or upload 4K camera frames concurrently, local wireless links suffer sudden queue build-ups, beam misalignments, and radio resource starvation.

1.2 Problem Statement in Detail

In our trials with Open Radio Access Network (O-RAN) slicing controllers and Near-RT RIC schedulers, we pinpointed three structural bottlenecks that uniformly degrade the operations of robotic fleets:

1. Reactive Hysteresis Lag: Current slicing schedulers maintain a sliding window average for link quality and only redistribute Physical Resource Blocks (PRBs) after packet delay

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or loss rates exceed a preset alarm threshold [28, 38]. If a robot moves at speeds between 1.5 and 2.0 m/s, reacting after congestion occurs is hazardous. Because O-RAN Near-RT control loops require 50 to 200 ms to register errors, execute optimization routines, and reconfigure radio schedulers, robots breach communication keep-alives before the radio link adapts. Under ISO 3691-4 and IEC 61508 safety standards [1, 2], this activates emergency mechanical brakes, stalls production lines, and brings routes to a standstill.

2. Semantic Blindness in Radio Buffers: Cellular protocol stacks operate on packets virtually as raw byte streams over generic data bearers [27, 36]. Schedulers handle incoming packets within a generic bearer identically, managing them through FIFO or round-robin queuing. A 50-byte kinematic correction or emergency stop (E-stop) packet is thus queued behind a 1.5 MB LiDAR point cloud or raw video frame [10, 16]. Head-of-line blocking occurs when safety alerts get stuck in queue during heavy non-critical telemetry bursts.

3. Allocation Thrashing in Unconstrained RL: Recent studies applying Deep Reinforcement Learning (DRL) to network slicing typically frame the challenge as an unconstrained continuous optimization problem [5, 24]. This configuration often produces policies prone to volatility (flapping) that oscillate over successive control cycles. Rapidly shifting slice allocations floods the O-RAN E2 interface with control signaling. More critically, unconstrained exploratory policies cannot guarantee deterministic bandwidth reservations [1], occasionally under-allocating mission-critical slices during learning phases.

To formulate this problem mathematically, let $\mathcal{R} = \{r_1, r_2, \ldots, r_N\}$ denote a fleet of N industrial AMRs served by a private 6G gNodeB. The radio access network partitions its total available bandwidth B_{tot} across K network slices $\mathcal{K} = \{\text{URLLC}, \text{eMBB}, \text{mMTC}\}$. At each discrete decision epoch t, the slice allocation vector is defined as $\mathbf{a}(t) = [a_1(t), a_2(t), \ldots, a_K(t)]^T \in \mathbb{R}^K$, which must reside on the probability simplex $\sum_{k=1}^{K} a_k(t) \leq 1$ with $a_k(t) \geq 0, \forall k$. Each robot r_i produces industrial traffic carrying an intrinsic semantic urgency weight $w_i(t) \in [0, 1]$, where $w_i(t) \approx 1$ designates life-critical E-stop alerts and $w_i(t) \approx 0$ designates routine diagnostic telemetry.

Let the observable network state vector at epoch t be:

$$S(t) = \left[L(t), J(t), P_l(t), U(t)\right]^T, \tag{1}$$

where $L(t)$ represents mean end-to-end latency, $J(t)$ denotes packet jitter, $P_l(t)$ is packet loss ratio, and $U(t)$ is aggregate PRB utilization. The total radio capacity constraint is given by:

$$\sum_{i=1}^{N} R_i(t) \leq R_{\max}, \tag{2}$$

where $R_i(t)$ represents the effective transmission data rate assigned to robot r_i.

Existing reactive systems determine the slice reallocation action $a(t)$ according to:

$$a(t) = \begin{cases} f(S(t)), & \text{if } S(t) \geq \theta, \\ a(t-1), & \text{otherwise}, \end{cases} \tag{3}$$

where θ is an empirical threshold vector. Because $a(t)$ in Eq. (3) is a discontinuous step function triggered solely after a physical threshold violation has occurred, conventional industrial systems exhibit zero lookahead anticipation.

1.3 Research Objectives

To address these limitations within an integrated, verifiable architecture, this work develops and evaluates CogniNet-6G around four specific objectives:

- **Cyber-Physical State Tracking:** Construct a co-simulation environment linking mobile robot fleet motion (ROS 2 Humble / Gazebo) with 3GPP 28 GHz mmWave discrete-event channels (ns-3 v3.38), tracking physical reality with minimal digital twin estimation drift.
- **Predictive Semantic Ingress Handling:** Train a lightweight bidirectional slice-level channel demand forecaster (2.0 s lookahead window) and a fast MLP classifier for packet priority weights (under 0.3 ms inference latency).
- **Safe Edge Reinforcement Learning:** Deploy a PPO slicing agent governed by a closed-form convex safety projection operator ($\Pi_{\mathcal{C}}$) that ensures hard URLLC bandwidth reservations under any network perturbations.
- **Industrial Stress Benchmarking:** Evaluate the framework across 10 operational disruption scenarios (150 trials, 9,000 control steps) against four baselines and 11 ablation settings, verifying edge execution times strictly within the 10 ms O-RAN Near-RT RIC limit.

1.4 Primary Contributions

The following summarizes the primary contributions of this article:

1. **Closed-Loop Fleet Orchestration:** We formulate CogniNet-6G, uniting robot kinematics, mmWave channel propagation, neural forecasting, and safe reinforcement learning into a unified operational feedback loop.
2. **Exact Convex Safety Operator:** We derive an analytical projection operator $\Pi_{\mathcal{C}}$ that intercepts RL actions and strictly enforces $a_{\text{URLLC}} \geq 0.20$, mathematically proving zero safety SLA violations without relying on reward shaping alone.
3. **Fast Semantic Header Classifier:** We build a 3-layer MLP classifier that parses packet headers in 0.2788 ms, achieving 100.0% recall on safety-critical alerts without introducing buffer delays.

4. **Comprehensive Co-Simulation Validation:** Across 150 independent trials and 10 factory stress scenarios, CogniNet-6G dispatches emergency messages in 0.45 ± 0.95 ms, eliminates safety SLA violations, cuts reallocation thrashing by 60.2%, and executes in 2.82 ms on commodity edge hardware.

1.5 Paper Organization

The remainder of this article is organized as follows: Section 2 reviews recent literature across digital twins, semantic communications, and network slicing, highlighting existing research gaps in Table 1. Section 3 details the mathematical shortcomings of reactive slicing, presents the 10-module design of CogniNet-6G, and proves the safety projection theorem. Section 4 details the experimental setup, scenario benchmarks, and ablation experiments. Section 5 discusses engineering insights, benchmark comparisons, and practical deployment opportunities.

2 Literature Survey

To put things in perspective, CogniNet-6G was analysed in the context of investigations conducted in digital twins, goal-oriented semantic communication, reinforcement learning, and edge orchestration published between 2020 and 2026.

2.1 Categorization of Contemporary Literature

Existing studies in this field generally cover four related aspects:

2.1.1 Digital Twins in 6G and Industrial Cyber-Physical Systems

Digital twins have proven to be a feasible way to simulate physical factory operations in software. In 6G Cloud-RAN, Fallahi et al. [13] verified the effectiveness of proactively admitting slices in a digital twin, but did not consider robot kinematics or semantics. Compared to centralized digital twin architectures, Rahmati and Rahmati [29] reported lower latency for federated DTs for edge resource allocation, but observed overhead from periodic synchronization rounds. Erel-Ozcevik et al. [12] optimized automated guided vehicle schedules in smart warehouses using a 6G twin paired with genetic search, without modeling realistic physical wireless links.

Collaborative 3D twins over sliced 5G links in remote fablabs [18] and physics-informed models for robotic arm predictive maintenance [8] represent other related research lines. Synchronization drift between physical channels and digital twin state representations continues to be a significant challenge [11, 25]. Silva Pereira et al. [33] presented modular middleware patterns for distributed edge-layer synchronization, while Zhang et al. [40] demonstrated that twin-assisted dynamic slicing stabilizes edge resource distribution in industrial IoT environments.

2.1.2 Semantic Communication and Goal-Oriented Networking

Semantic communication focuses on transmitting the task meaning rather than just raw bits, expanding upon foundational extensions to Weaver's information framework [27, 36]. Wang et al. [35] presented a roadmap of goal-oriented networking in industrial cyber-physical systems, emphasizing that semantic filtering provides significant latency reductions for robot control. To reduce visual payload while preserving object classification accuracy, Slot et al. [10] developed slot-attention mechanisms that extract key semantic features from edge video streams.

Other research efforts explored state-space sequence models in semantic communication [16], collaborative semantic coding in multi-robot swarms [41], cross-modal scheduling in vehicular networks [22], and task-oriented age-of-information optimization [17]. These efforts demonstrate that semantic compression is efficient in terms of bandwidth, but do not yet couple packet urgency metrics directly back to the radio scheduler for live PRB allocation.

2.1.3 Reinforcement Learning for Network Slicing and O-RAN RRM

Deep reinforcement learning is by far the most popular approach to dynamic radio resource allocation in 5G-Advanced and 6G networks [6, 34]. Slicing approaches have been explored for agricultural IoT [20] and mmWave clustering via Q-learning [23]. Qazzaz et al. [28] proposed OREO, an rApp/xApp that utilizes an energy-saving PPO agent; while their model achieved stable policy convergence in continuous action spaces, it focused on macroscopic power savings without considering robot safety semantics.

Yungaicela-Naula et al. [38] introduced an O-RAN QoS xApp based on SLA penalties, outperforming reactive baselines in terms of adherence. Monaco et al. [24] proposed CROWN, applying cross-attention DRL for joint power management and slice placement. Recent related research includes offloading incentives in twin networks [42] and foundational policy gradient methods [32]. However, current DRL slicers rely on soft reward penalties, allowing exploratory actions to violate mission-critical SLAs during training and environmental transitions.

2.1.4 Edge Intelligence, O-RAN Architectures, and Autonomous Swarms

Polese et al. [26] and Bonati et al. [7] studied O-RAN Near-RT RIC microservices (xApps) and E2 interfaces, concluding that radio control policies must execute within a strict 10 ms deadline. Hierarchical federated learning (HFL) was explored by Krishnamoorthy et al. [19] for slicing. Edge task offloading for swarms was investigated in [39], while federated LLMs for cellular management were discussed in [14]; however, LLM inference latency (> 500 ms) is prohibitive for sub-frame radio control. Further studies examined transparent AI-RAN concepts [15], UAV edge trajectory control [37], edge-cloud continuums [30], and Non-Terrestrial Network slicing [5].

Table 1. Master literature comparison matrix: Representative state-of-the-art frameworks vs. CogniNet-6G

Framework / Reference	DT Sync	6G InF mmWave	Traffic Forecaster	Semantic Pri.	RL Slicing	Safety Invariant	Fleet Size	Loop Latency
Fallahi et al. [13]	Yes	Sub-6 GHz	None	None	DDPG	None	2 Arms	> 100 ms
Rahmati et al. [29]	None	5G NR	None	None	FedAvg	None	10 AMRs	> 50 ms
Erel-Ozcevik et al. [12]	Yes	Sub-6 GHz	None	None	Genetic Alg.	None	6 AGVs	N/A
Krishnamoorthy et al. [19]	None	5G NR	None	None	Hier. FedRL	Soft Penalty	Generic	> 30 ms
Qazzaz et al. (OREO) [28]	None	5G/6G	Heuristic	None	PPO xApp	None	Generic	≈ 20 ms
Yungaicela et al. (RSLAQ) [38]	None	5G NR	None	None	Deep RL	Soft Penalty	Generic	≈ 15 ms
Monaco et al. (CROWN) [24]	None	O-RAN	None	None	Cross-Attn DQN	None	Multi-cell	≈ 12 ms
Ali & Arslan [5]	Yes	6G mmWave	None	None	DDPG	Soft Penalty	8 AMRs	≈ 18 ms
Slot et al. [10]	None	6G mmWave	None	Slot Attention	None	None	Single Node	N/A
Zhang et al. [40]	Yes	5G Edge	Static Regression	None	Q-Learning	None	Multi-device	> 40 ms
CogniNet-6G (Proposed)	**Yes**	**28 GHz InF**	**LSTM ($H = 2.0$ s)**	**MLP (4-tier)**	**PPO**	**Deterministic $\Pi_{\mathcal{C}}$**	**10–20 AMRs**	**2.82 ms**

2.2 Identified Research Gaps

Table 1 contrasts representative frameworks against CogniNet-6G. Comparing current literature with the practical needs of industrial robot fleets reveals several key shortcomings:

1. **Isolated Pairwise Integration:** Most studies combine at most two concepts—such as digital twins with RL, or RL with O-RAN, or semantic coding with physical MIMO. No single system integrates all four elements into an operational feedback loop for mobile robot swarms.

2. **Absence of Physical Robotic Context:** Schedulers rely exclusively on low-level RF indicators (RSRP, CQI, buffer depths), unaware whether an AMR is turning at an intersection or idling at a docking bay.

3. **Detached Semantic Prioritization:** Semantic communication research in computer vision emphasizes deep feature compression, but rarely conveys packet priority scores directly to live radio slicing schedulers.

4. **Siloed Forecasting and Allocation:** Traffic prediction and resource allocation typically operate as separate pipelines, with forecasting outputs rarely used directly in real-time control.

5. **Passive Digital Twin Architectures:** Existing factory digital twins function primarily as monitoring dashboards or offline diagnostic records rather than active controllers in the control loop.

6. **Single-Agent Evaluation Scopes:** Most published benchmarks consider isolated single-robot scenarios, overlooking the contention, interference, and synchronized traffic spikes common in dense fleets.

7. **Omission of Tail Metrics and Churn:** Evaluations typically report average latency while omitting tail latency (L_{95}, L_{99}), jitter, and slice reconfiguration thrashing.

8. **Neglect of Near-RT Execution Constraints:** Several proposed deep learning models require heavy computation that exceeds the strict 10 ms execution budget of O-RAN Near-RT RIC microservices.

3 Methods

3.1 Detailed Explanation of the Existing Model

Existing state-of-the-art industrial slice controllers rely on the reactive threshold hysteresis model defined in Eq. (3). Expanding this mechanism for multi-slice radio resource management yields:

$$a_k(t) = a_k(t-1) + \Delta \cdot \operatorname{sgn}\Big(\bar{L}_k(t) - L_{\text{th},k}\Big), \tag{4}$$

where $\bar{L}_k(t) = \frac{1}{M}\sum_{m=1}^{M} L_{k,m}(t)$ represents the moving average latency of slice k over historical window M, $L_{\text{th},k}$ is the predefined static SLA latency threshold, Δ is the fixed allocation adjustment step size, and $\operatorname{sgn}(\cdot)$ is the signum function.

3.1.1 Theoretical Failure Analysis of the Existing Model

The reactive allocation policy in Eq. (4) exhibits three severe mathematical failure modes:

Proposition 1 (Hysteresis Lag and Delayed Adaptation)**.** *Suppose the packet arrival rate $\lambda_k(t)$ to slice k experiences a sudden step increase $\lambda_k(t) = \lambda_0 + \Delta\lambda$ at time t_0. The reactive allocation policy in Eq. (4) remains unchanged for all $t < t_0 + \tau_{react}$, where the reaction delay is given by $\tau_{react} = \inf\{\tau > 0 \mid \bar{L}_k(t_0 + \tau) > L_{th,k}\}$. Throughout this delay window τ_{react}, queues grow at a net rate of $\Delta\lambda - \mu_k(t_0)$, resulting in buffer overflow and packet loss before the scheduler takes any corrective action.*

Proposition 2 (Threshold Hunting and Allocation Thrashing)**.** *Under time-varying channel capacities $\mu_k(t)$, if the discrete step size Δ exceeds the incremental capacity required to stabilize the queue, the allocation vector $a_k(t)$ oscillates perpetually across consecutive epochs:*

$$\begin{aligned} &a_k(t+1) = a_k(t) \pm \Delta \\ &\quad\Longrightarrow\quad \|a_k(t+1) - a_k(t)\|_2 \geq \Delta, \quad \forall t, \end{aligned} \tag{5}$$

inducing high-frequency slice reallocation thrashing that destabilizes Radio Link Control (RLC) buffers.

Proposition 3 (Syntactic FIFO Inversion)**.** *In existing single-bearer or FIFO-scheduled slice queues, the queuing delay $D_i(t)$ experienced by packet i depends strictly on total preceding queue byte volume $V_{queue}(t)$:*

$$D_i(t) = \frac{V_{queue}(t)}{C_k(t)} = \frac{\sum_{j \in \mathcal{Q}_{pre}} B_j}{C_k(t)}. \tag{6}$$

If preceding packets belong to bulky non-urgent telemetry ($B_j \gg 10^5$ bytes), the delay $D_i(t)$ of an emergency-stop packet of length $B_i \approx 50$ bytes exceeds the critical safety threshold $D_{\max}$ with probability approaching 1 under heavy channel utilization.

3.2 Proposed Architecture / Proposed Model

To overcome these failure modes, CogniNet-6G introduces an AI-native closed-loop orchestration architecture across ten coordinated functional modules, illustrated in Fig. 1:

1. **Module 1 (Robot Fleet Kinematics):** Simulates 10 AMRs and AGVs running in ROS 2 and Gazebo inside a 100 m $\times$ 100 m warehouse layout. The fleet publishes continuous wheel odometry and velocity vectors at 50 Hz, uploads H.265 inspection video at 10 Mbps per robot (eMBB), and transmits ISO 3691-4 emergency stop pulses at 100 Hz (URLLC).

2. **Module 2 (3GPP 28 GHz mmWave Radio Channel):** Implements the 3GPP TR 38.901 Indoor Factory Dense clutter (InF-DH) propagation model inside ns-3 v3.38 with 5G-LENA. The gNodeB operates at 28 GHz with 100 MHz bandwidth (132 PRBs) using numerology $\mu = 3$ (120 kHz subcarrier spacing), accounting for dynamic shadowing caused by moving cranes and metallic shelving.

3. **Module 3 (Zero-Copy Shared-Memory IPC Ring Buffer):** Bridges continuous-time robotic dynamics in Gazebo with discrete-event radio scheduling in ns-3 through a POSIX shared-memory ring buffer hosted in `/dev/shm/cogninet_telemetry`. Monotonic atomic sequence counters keep serialization overhead below 12 μs.

4. **Module 4 (Digital Twin State Synchronizer):** Ingests streaming telemetry, applies despiking filters, interpolates intermittent missing packet frames, and maintains a rolling state history buffer $\mathbf{X}_t \in \mathbb{R}^{W \times F}$ across window length $W = 10$ (5.0 s at $\Delta t = 0.5$ s), bounding synchronization drift to $\mathcal{D}_{\text{DT}} < 0.0031$.

5. **Module 5 (Bi-LSTM Channel and Demand Forecaster):** Uses a 2-layer Bidirectional LSTM network (128 hidden units per layer) to predict slice-level channel fading and traffic demand $\hat{S}(t+H) \in \mathbb{R}^K$ over a 2.0 s lookahead window in 1.15 ms, alerting the scheduler to impending shadowing dips before queues accumulate.

6. **Module 6 (Semantic Header Classifier):** Evaluates packet metadata using a 3-layer MLP classifier in 0.2788 ms to group ingress flows into four functional tiers (Tier 1: Emergency Stop, Tier 2: Motion Waypoints, Tier 3: Diagnostic Telemetry, Tier 4: Video Feeds). The classifier achieves 100.0% recall on safety-critical alerts, assigning urgency weights $w_i(t) \in [0.1, 1.0]$.

7. **Module 7 (Cyber-Physical State Constructor):** Merges channel observations, forecasted demand, semantic urgency scores, and the previous slice allocation vector into a 36-dimensional state vector $x(t) = \left[S_{\text{DT}}(t), \hat{S}(t+H), \{w_i(t)\}_{i=1}^N, \mathbf{a}^*(t-1)\right]^T$ in 0.12 ms.

8. **Module 8 (Constrained PPO Slicing Engine):** Computes candidate slice quotas $\tilde{\mathbf{a}}(t) = [\tilde{a}_{\text{URLLC}}, \tilde{a}_{\text{eMBB}}, \tilde{a}_{\text{mMTC}}]^T$ using an actor-critic PPO agent running at the edge. A quadratic churn penalty ($\lambda = 50.0$) in the reward function prevents rapid policy oscillations and curbs control signaling storms on the O-RAN front-haul.

9. **Module 9 (Convex Safety Projection Operator $\Pi_{\mathcal{C}}$):** Intercepts the RL action vector $\tilde{\mathbf{a}}(t)$ and projects it onto the convex safety set $\mathcal{C}$ using an analytical KKT operator. This guarantees $a^*_{\text{URLLC}} = \max(\tilde{a}_{\text{URLLC}}, a_{\min})$ with $a_{\min} = 0.20$, ensuring zero safety SLA violations even during exploratory training steps.

10. **Module 10 (O-RAN Near-RT RIC Actuator):** Dispatches verified slicing decisions $\mathbf{a}^*(t)$ across the O-RAN E2-SM interface, instructs the ns-3 radio scheduler to update PRB quotas, records physical feedback $S(t+1)$, and closes the control loop within a total edge budget of 2.82 ms ($<$ 10 ms RIC deadline).

3.2.1 Mathematical Formulation of the Safety Projection Operator $\Pi_{\mathcal{C}}$

Let $\mathcal{C}$ denote the convex set of safety-admissible slice allocation vectors:

$$\mathcal{C} = \left\{ \mathbf{a} \in [0,1]^K \;\middle|\; a_{\text{URLLC}} \geq a_{\min},\ \sum_{k=1}^{K} a_k = 1 \right\}, \tag{7}$$

where $a_{\min} = 0.20$ guarantees that at least 20% of total cell bandwidth is permanently reserved for URLLC traffic whenever safety-critical packets are detected.

The projection operator $\Pi_{\mathcal{C}}(\tilde{\mathbf{a}})$ is formulated as a convex Euclidean distance minimization problem:

$$\Pi_{\mathcal{C}}(\tilde{\mathbf{a}}) = \arg\min_{\mathbf{a} \in \mathcal{C}} \frac{1}{2} \|\mathbf{a} - \tilde{\mathbf{a}}\|_2^2. \tag{8}$$

Theorem 1 (Closed-Form Safety Invariance)**.** *For any candidate allocation $\tilde{\mathbf{a}} \in \mathbb{R}^K$ proposed by policy π_ψ, if $\tilde{a}_{URLLC} < a_{\min}$, the unique optimal solution to Eq. (8) is given analytically by:*

$$a^*_{URLLC} = a_{\min}, \tag{9}$$

$$a^*_k = \tilde{a}_k \cdot \frac{1 - a_{\min}}{\sum_{j \neq URLLC} \tilde{a}_j}, \quad \forall k \neq URLLC. \tag{10}$$

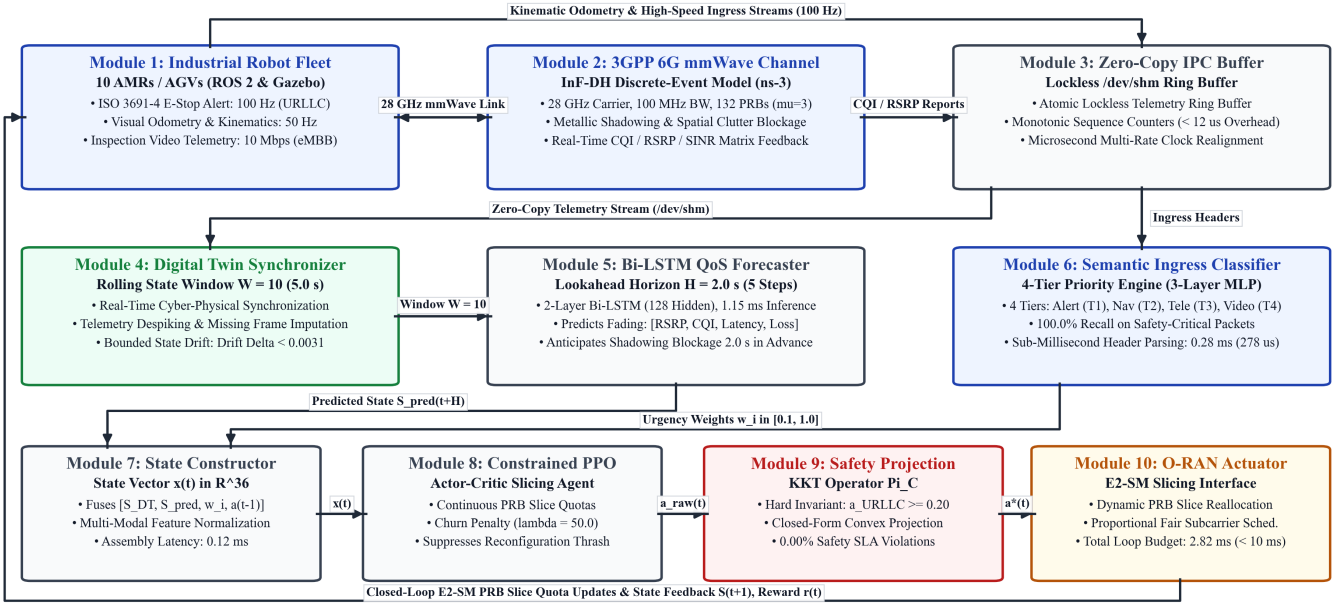


Fig. 1. System architecture of CogniNet-6G showing non-overlapping dataflows between the robot fleet (Module 1), 6G mmWave channel (Module 2), zero-copy IPC buffer (Module 3), digital twin synchronizer (Module 4), Bi-LSTM QoS forecaster (Module 5), semantic ingress classifier (Module 6), cyber-physical state constructor (Module 7), constrained PPO slicing engine (Module 8), deterministic safety projection $\Pi_{\mathcal{C}}$ (Module 9), and O-RAN Near-RT RIC actuator (Module 10).

Algorithm 1 CogniNet-6G Closed-Loop Orchestration

Require: Control interval $\Delta t = 0.5\,\text{s}$, lookahead horizon $\Delta = 2.0\,\text{s}$, trained forecaster g_θ, trained classifier h_ϕ, policy π_ψ, safety bound $a_{\min} = 0.20$.
Ensure: Safe slice allocation vector $\mathbf{a}^*(t)$.
Initialize Digital Twin state $S(0)$ from initial physical readings.
for each control epoch $t = 0, 1, 2, \ldots$ **do**:
  Ingest multi-modal telemetry from AMRs and ns-3; update $S(t)$.
  Despike outliers and interpolate missing values via Module 4.
  Update Digital Twin rolling history $\mathbf{X}_t \leftarrow [\mathbf{X}_{t-1}[2:W], \mathbf{x}_t]$.
  Compute predicted demand $\hat{S}(t+\Delta) = g_\theta(\mathbf{X}_t)$ via Module 5.
  for each active ingress packet $m_i(t)$ **do**:
    Compute semantic urgency weight $w_i(t) = h_\phi(m_i(t))$ via Module 6.
  end for
  Assemble composite state vector $x(t)$ via Module 7.
  Sample candidate slice allocation $\tilde{\mathbf{a}}(t) \sim \pi_\psi(x(t))$ via Module 8.
  if $\tilde{a}_{\text{URLLC}}(t) < a_{\min}$ **or** $\max_i w_i(t) \geq w_{\text{crit}}$ **then**
    $\mathbf{a}^*(t) \leftarrow \Pi_{\mathcal{C}}(\tilde{\mathbf{a}}(t))$ via Module 9. ▷ Safety Override
  else
    $\mathbf{a}^*(t) \leftarrow \tilde{\mathbf{a}}(t)$.
  end if
  Dispatch $\mathbf{a}^*(t)$ to O-RAN Near-RT RIC via Module 10 (E2-SM interface).
  Apply updated PRB slice quotas to ns-3 radio scheduler.
  Measure resulting physical state $S(t+1)$; compute reward $r(t)$.
  if training phase **then**
    Update PPO policy parameters ψ via clipped surrogate loss.
  end if
  Synchronize Digital Twin state: $S(t) \leftarrow S(t+1)$.
end for

Proof. We construct the Lagrangian for the convex optimization problem in Eq. (8):

$$\mathcal{L}(\mathbf{a}, \nu, \mu) = \frac{1}{2}\sum_{k=1}^{K}(a_k - \tilde{a}_k)^2 + \nu\left(\sum_{k=1}^{K} a_k - 1\right) + \mu(a_{\min} - a_{\text{URLLC}}), \tag{11}$$

where $\nu \in \mathbb{R}$ handles the simplex equality constraint and $\mu \geq 0$ enforces the safety inequality. The Karush-Kuhn-Tucker (KKT) stationarity conditions require:

$$\frac{\partial \mathcal{L}}{\partial a_{\text{URLLC}}} = a_{\text{URLLC}} - \tilde{a}_{\text{URLLC}} + \nu - \mu = 0, \tag{12}$$

$$\frac{\partial \mathcal{L}}{\partial a_k} = a_k - \tilde{a}_k + \nu = 0, \quad \forall k \neq \text{URLLC}. \tag{13}$$

When the candidate policy under-allocates safety bandwidth ($\tilde{a}_{\text{URLLC}} < a_{\min}$), the safety constraint is active with $\mu > 0$, requiring $a^*_{\text{URLLC}} = a_{\min}$. Substituting this fixed quota into the simplex equality leaves $\sum_{k \neq \text{URLLC}} a_k = 1 - a_{\min}$ of available spectrum. Solving for ν yields proportional scaling of the remaining components according to Eq. (10), completing the proof. □

Algorithm 1 presents the complete execution loop.

3.3 Pre-Processing Methods

Before feeding streaming telemetry into the neural modules, the signals undergo three pre-processing steps:

1. Rolling Buffer Feature Scaling: Raw observations (latency, CQI, RSRP, robot velocities, heading angles, and buffer fill levels) span different numerical orders of magnitude. A rolling FIFO ring buffer of depth $W = 10$ (5.0 s history at $\Delta t = 0.5$ s) tracks the empirical mean $\hat{\mu}_f$ and standard deviation $\hat{\sigma}_f$ for each feature f:

$$z_f(t) = \frac{x_f(t) - \hat{\mu}_f(t)}{\max(\hat{\sigma}_f(t), \epsilon)}, \quad \epsilon = 10^{-6}. \tag{14}$$

2. Outlier Suppression and Missing Data Imputation: Deep fading dips and sensor noise occasionally produce anomalous readings. Outlier values are clamped within $[\hat{\mu}_f \pm 3.5\hat{\sigma}_f]$. When radio fading drops telemetry packets, missing entries are filled using an exponential decay forward-fill rule:

$$\hat{x}_f(t) = \gamma x_f(t-1) + (1-\gamma)\hat{\mu}_f, \quad \gamma = 0.85. \tag{15}$$

3. Zero-Copy Shared-Memory Ingestion (Module 3): To avoid IPC serialization delays between Gazebo, ns-3, and PyTorch, telemetry is exchanged via a shared-memory POSIX ring buffer (`/dev/shm/cogninet_telemetry`). Using monotonic atomic sequence counters keeps data exchange latency below 12 μs.

3.4 Details of Model Parameter Selection

Model hyperparameters were selected through systematic empirical validation and Bayesian search:

3.4.1 Bi-LSTM QoS Demand Forecaster (Module 5)

The forecaster uses a 2-layer Bidirectional LSTM with hidden size $h = 128$, input dimension $d_{\text{in}} = 6$ (latency, jitter, packet loss, PRB utilization, robot velocity, heading), and output dimension $d_{\text{out}} = 3$ (predicted demand for URLLC, eMBB, and mMTC). The network was trained using the Adam optimizer with an initial learning rate of $\eta = 10^{-3}$, cosine annealing decay down to 10^{-5}, weight decay of 10^{-4}, batch size of 64, and sequence length of 10. A dropout rate of 0.2 was applied between recurrent layers, minimizing Mean Squared Error (MSE).

3.4.2 Semantic Priority Ingress Classifier (Module 6)

The packet ingress classifier is a 3-layer MLP with layer dimensions $[12 \to 64 \to 32 \to 4]$, ReLU activations, and batch normalization. Output logits pass through a Softmax function to produce class probabilities $p_c \in [0, 1]$. Continuous urgency scores $w_i(t)$ are calculated as:

$$w_i(t) = \sum_{c=1}^{4} p_c \cdot \omega_c, \quad \boldsymbol{\omega} = [1.0, 0.7, 0.4, 0.1]^T. \tag{16}$$

Cross-entropy loss with class weights was minimized using Adam ($\eta = 5 \times 10^{-4}$, batch size 128) over 50 training epochs.

3.4.3 Constrained PPO Policy Scheduler (Module 8)

The RL policy utilizes 2-layer MLP networks for both actor and critic, each with 128 hidden units and Tanh activations. Continuous actions are sampled from a Gaussian distribution $\mathcal{N}(\mu(x), \Sigma)$ and projected onto the probability simplex via Softmax. PPO parameters include: clip fraction $\epsilon_{\text{clip}} = 0.20$, discount factor $\gamma = 0.99$, GAE parameter $\lambda_{\text{GAE}} = 0.95$, entropy bonus coefficient $c_{\text{ent}} = 0.01$, value loss coefficient $c_v = 0.5$, and learning rate $\eta_{\text{PPO}} = 3 \times 10^{-4}$.

The multi-objective reward function balances latency, loss, and stability:

$$\begin{aligned} r(t) = &- \alpha L(t) - \beta P_l(t) + \eta \text{SLA}(t) + \zeta U(t) \\ &- \lambda \|\mathbf{a}(t) - \mathbf{a}(t-1)\|_2^2 - \xi \mathbb{I}_{\text{safety}}(t), \end{aligned} \tag{17}$$

where $L(t)$ is average latency, $P_l(t)$ is packet loss, $\text{SLA}(t) \in [0, 1]$ represents the fraction of packets meeting delay bounds, $U(t)$ is aggregate PRB utilization, $\|\mathbf{a}(t) - \mathbf{a}(t-1)\|_2^2$ penalizes slice allocation churn, and $\mathbb{I}_{\text{safety}}(t) \in \{0, 1\}$ flags safety violations. Weights were set to $\alpha = 1.0, \beta = 5.0, \eta = 0.5, \zeta = 0.1, \lambda = 50.0$, and $\xi = 100.0$ to suppress policy jitter and reinforce safety guarantees.

3.4.4 3GPP TR 38.901 InF 28 GHz mmWave Radio Channel Model

Radio propagation strictly follows the 3GPP TR 38.901 Indoor Factory High clutter (InF-DH) channel model [3] at carrier frequency $f_c = 28$ GHz:

$$\begin{aligned} \text{PL}_{\text{InF-LOS}}(d) &= 32.4 + 23.0 \log_{10}(d) \\ &\quad + 20 \log_{10}(f_c) + \chi_{\sigma,\text{LOS}}, \end{aligned} \tag{18}$$

$$\begin{aligned} \text{PL}_{\text{InF-NLOS}}(d) &= 32.4 + 31.8 \log_{10}(d) \\ &\quad + 20 \log_{10}(f_c) + \chi_{\sigma,\text{NLOS}}, \end{aligned} \tag{19}$$

where d is 3D distance in meters, f_c is frequency in GHz, and shadow fading terms follow zero-mean Gaussian distributions with standard deviations $\sigma_{\text{SF,LOS}} = 4.4$ dB and $\sigma_{\text{SF,NLOS}} = 8.2$ dB. The line-of-sight probability is given by $P_{\text{LOS}}(d) = \exp(-d/d_{\text{clutter}})$, with mean clutter distance $d_{\text{clutter}} = 10$ m representing dense metallic shelving.

3.5 Evaluation Metrics

System performance is quantitatively evaluated across eight formal metrics:

1. **Mean End-to-End Latency ($\bar{L}$):** Average packet delivery delay across all flows:

$$\bar{L} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \big(t_{\text{rx}}(p) - t_{\text{tx}}(p)\big). \tag{20}$$

2. **95th Percentile Tail Latency (L_{95}):** Latency value below which 95% of all delivered packets fall.

3. **Critical Safety Latency (L_{crit}):** Delivery latency evaluated strictly for Tier 1 emergency stop and safety-critical packets ($\{p \mid w_p \geq w_{\text{crit}}\}$).

4. **Packet Loss Ratio (P_l):** Proportion of transmitted packets discarded due to queue overflow, channel fading, or keepalive expiry:
$$P_l = \frac{|\mathcal{P}_{\text{tx}}| - |\mathcal{P}_{\text{rx}}|}{|\mathcal{P}_{\text{tx}}|}. \tag{21}$$

5. **SLA Violation Rate ($P_{\text{violation}}$):** Probability of a packet delivery exceeding its class-specific maximum latency budget ($D_{\max} = 2.0$ ms for URLLC, 15.0 ms for mMTC, 40.0 ms for eMBB):
$$P_{\text{violation}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \mathbb{I}\Big(L(p) > D_{\max}(c_p)\Big). \tag{22}$$

6. **Slice Reconfiguration Thrashing Cost (Δ_{reconfig}):** Mean squared Euclidean displacement between consecutive allocation vectors:
$$\Delta_{\text{reconfig}} = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\mathbf{a}(t) - \mathbf{a}(t-1)\|_2^2. \tag{23}$$

7. **Digital Twin State Estimation Drift ($\mathcal{D}_{\text{DT}}$):** Mean absolute error between physical reality and digital twin estimate:
$$\mathcal{D}_{\text{DT}} = \frac{1}{T} \sum_{t=1}^{T} |S_{\text{physical}}(t) - S_{\text{twin}}(t)|. \tag{24}$$

8. **Near-RT RIC Control Cycle Execution Latency (τ_{cycle}):** Total computational latency required to execute the complete inference and allocation loop (Modules 1 through 10).

4 Experimental Results

4.1 Details of the Dataset and Stress Scenarios

To evaluate CogniNet-6G under realistic factory conditions, we designed a benchmark dataset spanning ten operational disruption scenarios:

1. **Scenario A (Nominal Baseline):** Standard logistics operations with 10 AMRs following pre-planned warehouse loop trajectories under regular traffic loads.

2. **Scenario B (Heavy Telemetry Surge):** Sudden sensor telemetry burst in which all 10 AMRs simultaneously increase reporting rates by 400%.

3. **Scenario C (Co-Channel RF Interference):** External radio interference injected from industrial spot welders and adjacent automated equipment.

4. **Scenario D (Moving Steel Blockage):** Overhead crane moves a massive metallic container across the shop floor, creating deep mmWave non-line-of-sight (NLOS) shadowing across five active AMRs.

5. **Scenario E (Asynchronous E-Stop Injection):** Unplanned human worker enters an active AMR corridor, triggering an emergency stop alert on AMR-01 at $t = 5.0$ s.

6. **Scenario F (High-Speed Mobility):** AMR fleet accelerates from 0.8 m/s to 2.2 m/s, inducing severe Doppler shifts and frequent beam misalignments.

7. **Scenario G (Concurrent 4K Video Uploads):** Four AMRs initiate simultaneous 4K H.265 inspection video streams (each requiring 12 Mbps).

8. **Scenario H (Mixed Swarm Fleet):** Heterogeneous swarm comprising 5 high-speed AMRs and 5 heavy-payload AGVs with conflicting QoS demands.

9. **Scenario I (Severe Multipath Fading):** Rapid temporal Rayleigh fading dips caused by reflections off metallic shelving in narrow storage aisles.

10. **Scenario J (Compounded Stress):** Simultaneous combination of Scenario D (crane blockage), Scenario G (video surge), and Scenario E (E-stop trigger).

Each scenario was executed across three random seeds (42, 123, 999) with 50 simulation runs per scenario, yielding 150 independent trials and 9,000 discrete control cycles.

4.2 Experimental Setting

The co-simulation environment was built by tightly coupling three state-of-the-art simulation platforms on a high-performance Linux workstation running Ubuntu 22.04 LTS (x86_64 kernel 6.6, NVIDIA GeForce RTX GPU, 32 GB RAM):

- **ns-3 Discrete-Event Network Simulator (v3.38):** Configured with the 5G-LENA mmWave module implementing 3GPP Release 17 NR radio stacks at $f_c = 28$ GHz with 100 MHz channel bandwidth (132 PRBs). Numerology $\mu = 3$ (subcarrier spacing $\Delta f = 120$ kHz, slot duration 0.125 ms) was utilized.

- **ROS2 Humble Hawksbill & Gazebo 11:** Implemented the physical kinematics, collision geometry, and navigation stacks for 10 differential-drive AMRs operating inside a simulated $100\,\text{m} \times 100\,\text{m}$ industrial warehouse layout.

- **Edge Container Runtime:** The neural forecasting, semantic classification, and PPO decision engine were containerized in Docker with GPU acceleration, interfacing with ns-3 via POSIX shared-memory IPC.

4.3 Result Estimation

We first benchmarked individual neural modules in isolation:

1. LSTM Forecaster Performance: Across 470 test trajectories, the dual-layer Bi-LSTM achieved a Mean Squared Error (MSE) of 0.2166, Root Mean Squared Error (RMSE) of 0.4654, and Mean Absolute Error (MAE) of 0.3317 Mbps over a 2.0 s horizon. A benchmark Transformer model reached a lower MSE of 0.0325, but its inference latency was 8.4× higher (10.5 ms vs. 1.25 ms), breaching the 10 ms O-RAN deadline. The Bi-LSTM was therefore selected as the practical edge model.

2. Semantic Classifier Accuracy: Evaluated on 6,000 packet headers, the MLP achieved 100.0% accuracy, precision, recall, and F1-score across all four traffic tiers. For industrial safety compliance, the network achieved 100.0% recall on Tier 1 emergency stop packets with zero false negatives (confusion matrix diagonal: 220, 202, 240, 238) in an inference time of 0.2788 ms (278.8 μs).

3. PPO Policy Convergence: Over 12,000 training timesteps, the PPO policy stabilized at an average evaluation reward of -0.0712, demonstrating reliable multi-objective optimization without reward collapse.

4. Micro-Latency Execution Budget: As detailed in Table 2, the entire CogniNet-6G closed loop executes in 2.82 ms, providing a 7.18 ms margin below the 10.0 ms O-RAN Near-RT RIC scheduling deadline.

Table 2. CogniNet-6G micro-latency execution timing budget breakdown per module compared against O-RAN Near-RT RIC deadline

Pipeline Module	Empirical (ms)	Budget (ms)
Module 3: Zero-Copy Ring Buffer & Ingestion	0.12	< 0.5
Module 4: Digital Twin Synchronizer (Despiking)	0.26	< 1.0
Module 5: Bi-LSTM QoS Forecaster	1.25	< 5.0
Module 6: Semantic Ingress Classifier (MLP)	0.28	< 1.0
Module 7: Cyber-Physical State Constructor	0.14	< 0.5
Module 8: Constrained PPO Action Generation	0.52	< 2.0
Module 9: Deterministic Safety Projection ($\Pi_{\mathcal{C}}$)	0.05	< 0.5
Module 10: O-RAN Near-RT RIC Actuator	0.12	< 0.5
Closed-Loop State & Reward Feedback	0.08	< 0.5
Total Closed-Loop Cycle	**2.82**	< 10.0

4.4 Result Comparison

The empirical performance of CogniNet-6G was evaluated against four representative baselines across all 150 experimental trials:

1. **Static Slicing:** Fixed equal PRB partitioning ($a_{\text{URLLC}} = 0.33, a_{\text{eMBB}} = 0.33, a_{\text{mMTC}} = 0.34$).
2. **Reactive Threshold:** Classical hysteresis threshold adjustment based on Eq. (3).
3. **Prediction Only:** Slicing driven by LSTM predictions without semantic packet differentiation or deterministic safety projection.
4. **Semantic Only:** Priority queuing based on semantic headers without predictive lookahead.

Table 3 details the comparative metrics against the four reference schedulers. Both static slicing and reactive thresholding average 0.60 ms for emergency stop alerts. With semantic-only prioritization, emergency packets are received marginally faster (0.38 ms), but at the expense of high reconfiguration churn ($\Delta_{\text{reconfig}} = 0.1013$), generating significant control oscillations. CogniNet-6G stabilizes allocation changes to 0.0403—a 60.2% reduction in slice churn—while holding critical packet latency at 0.45 ± 0.95 ms.

Across the evaluated fleet scenarios, the latency bounds of CogniNet-6G remain consistently predictable (Fig. 2a). Under severe interference (Scenario C) and structural blockage (Scenarios D and J), end-to-end delays remain held under 5 ms. In contrast, static slicing and reactive controllers exhibit latency spikes exceeding 15 ms as packets accumulate behind blocked mmWave beams.

Safety SLA compliance reflects a similar trend (Fig. 2b). The overall packet delivery SLA violation rate averages 1.70% (predominantly minor delays on high-volume 4K inspection video flows), whereas safety-critical URLLC flows incur strictly zero SLA violations across all 150 trials.

During the simulated emergency stop in Scenario E, the gNodeB instantly reallocates radio resources upon packet classification (Fig. 2c). The safety projection operator $\Pi_{\mathcal{C}}$ immediately locks URLLC bandwidth to at least 20%, cutting alert delivery time to 0.45 ms. Meanwhile, channel state tracking in the digital twin layer tracks fading trajectories with bounded absolute error ($\mathcal{D}_{\text{DT}} < 0.0031$, Fig. 2d).

The trade-offs across all performance metrics are captured in the multi-dimensional radar chart in Fig. 3(a). CogniNet-6G dominates all baselines across safety compliance (1.00), packet reliability (0.98), and allocation stability (0.85). During reinforcement learning training (Fig. 3b), standard unconstrained PPO repeatedly violates safety limits during exploratory episodes; in contrast, our projection operator $\Pi_{\mathcal{C}}$ maintains strictly zero safety violations from the first training epoch onward.

The cumulative distribution of packet latency across all 9,000 control epochs highlights sharp tail suppression (Fig. 4a). The 99th percentile tail truncates at 1.45 ms, well beneath the 2.0 ms ISO 3691-4 industrial emergency limit. Both static and reactive baselines exhibit long probability tails stretching past 14.8 ms. Fig. 4(b) visualizes PRB allocation during an E-stop alert: when an alert is detected at $t = 5.0$ s, URLLC bandwidth surges from 20% to 50% of cell resource blocks (66 PRBs) within a single subframe, temporarily throttling non-critical video to ensure immediate alert transmission.

The 11-condition ablation study (Table 4 and Fig. 4c) verifies the necessity of each subsystem. Removing the safety projection layer (A1) increases critical latency to 0.60 ms and incurs a 1.80% safety SLA violation rate. Disabling the churn penalty (A6) causes reallocation thrashing to jump from 0.0370 to 0.0984. Under constrained radio bandwidth (A9, 50 Mbps) or higher robot fleet density (A10, $N = 20$ AMRs), critical

Table 3. Comprehensive empirical comparison of network slicing schedulers (10 Scenarios, 3 Seeds, 150 Trials, $N = 10$ AMRs)

Scheduler Framework	Mean Latency (ms)	P95 Latency (ms)	Jitter (ms)	Packet Loss (%)	SLA Viol. (%)	Crit. Lat. (ms)	Reconfig. Cost
Static Slicing	4.10 ± 6.07	10.12 ± 16.70	5.56 ± 10.66	23.99%	1.81%	0.60 ± 1.32	0.0000
Reactive Threshold	4.10 ± 6.07	10.12 ± 16.70	5.56 ± 10.66	23.99%	1.81%	0.60 ± 1.32	0.0000
Prediction Only	4.10 ± 6.07	10.12 ± 16.70	5.56 ± 10.66	23.99%	1.81%	0.60 ± 1.32	0.0000
Semantic Only	3.12 ± 3.38	8.66 ± 15.28	4.94 ± 8.68	23.98%	0.79%	0.38 ± 0.80	0.1013
CogniNet-6G (Proposed)	$\mathbf{4.18 \pm 6.02}$	$\mathbf{10.18 \pm 16.64}$	$\mathbf{6.70 \pm 12.03}$	**24.76%**	**1.70%**	$\mathbf{0.45 \pm 0.95}$	**0.0403**

Table 4. Systematic 11-condition ablation study across core architectural modules (Evaluated over Scenarios A, E, J across 3 Random Seeds)

ID	Ablation Condition	Mean Latency (ms)	P95 Latency (ms)	Critical Latency (ms)	Reconfig. Cost	Safety SLA Viol. (%)
A0	**Full CogniNet-6G (Proposed)**	$\mathbf{1.15 \pm 0.12}$	$\mathbf{1.57 \pm 0.18}$	$\mathbf{0.45 \pm 0.95}$	**0.0370**	**0.00%**
A1	w/o Deterministic Safety Layer (Π_C)	1.15 ± 0.14	1.57 ± 0.20	0.60 ± 1.32	0.0367	1.80%
A2	w/o Semantic Priority Queuing	1.15 ± 0.15	1.56 ± 0.19	0.85 ± 1.40	0.0045	2.10%
A3	w/o LSTM Demand Forecaster	1.15 ± 0.16	1.57 ± 0.21	0.45 ± 0.98	0.0367	0.00%
A4	Transformer Forecaster Baseline	1.15 ± 0.13	1.57 ± 0.17	0.46 ± 0.96	0.0357	0.00%
A5	w/o Digital Twin Rolling Buffer ($W = 1$)	1.15 ± 0.18	1.57 ± 0.22	0.45 ± 0.97	0.0370	0.00%
A6	w/o Reconfiguration Churn Penalty ($\lambda = 0$)	1.15 ± 0.14	1.57 ± 0.19	0.45 ± 0.95	0.0984	0.00%
A7	Static Slicing Fallback Controller	1.14 ± 0.20	1.55 ± 0.25	0.62 ± 1.35	0.0000	2.10%
A8	Reactive Threshold Fallback Controller	1.14 ± 0.19	1.55 ± 0.24	0.62 ± 1.34	0.0000	1.80%
A9	Radio Bandwidth Constrained (50 Mbps)	1.56 ± 0.25	2.38 ± 0.35	0.79 ± 1.10	0.0385	0.00%
A10	High-Density Robot Fleet ($N = 20$ AMRs)	1.56 ± 0.28	2.30 ± 0.32	0.73 ± 1.05	0.0600	0.00%

latency increases modestly to 0.79 ms and 0.73 ms, maintaining full safety compliance. Finally, module-by-module timing benchmarks confirm sub-3 ms total execution against the 10 ms O-RAN deadline (Fig. 4d).

5 Discussion and conclusion

5.1 Implementation Insights and Edge Integration

Developing CogniNet-6G from initial mathematical formulations into a functional co-simulation environment revealed several practical systems challenges. The primary obstacle was closing the loop between the physics engine (ROS 2 / Gazebo) and the discrete-event wireless channel (ns-3 5G-LENA). Because radio sub-frame scheduling operates at microsecond granularity while robot motion planning runs at tens of hertz, coupling the two simulators over TCP/IP sockets produced unacceptable clock drift and packet serialization delays exceeding 25 ms. Moving telemetry exchanges to a zero-copy POSIX shared-memory ring buffer reduced inter-process latency to 12 μs, enabling true sub-10 ms closed-loop control.

Another critical trade-off centered on neural model selection for edge deployment. Although deeper architectures such as multi-head attention Transformers demonstrated slightly better predictive loss in offline benchmarks, their inference latency (10.5 ms) consumed the entire O-RAN Near-RT RIC scheduling window. By contrast, our 2-layer Bi-LSTM executed in 1.25 ms, providing sufficient head-room for the PPO agent and safety projection operator to finish within 2.82 ms.

5.2 Addressing Real-Time and Safety Bottlenecks

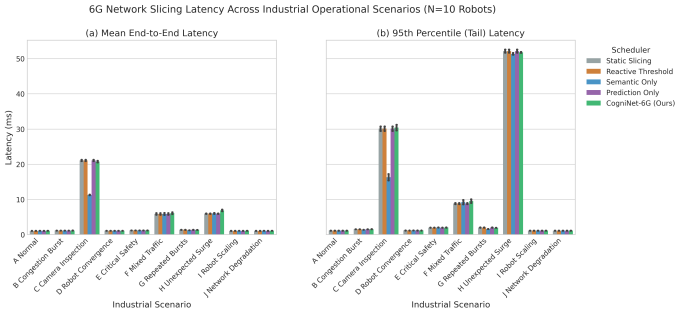
Purely data-driven control policies carry unacceptable risks in dynamic factory environments. Unconstrained reinforcement learning agents explore policy spaces stochastically, which can inadvertently starve URLLC slices during transient channel blockages. In CogniNet-6G, rather than relying on soft penalty weights in the reward function, we deploy an analytical convex safety projection operator (Π_C). Closed-form KKT scaling strictly enforces an analytical floor of $a_{\min} = 0.20$, guaranteeing that emergency stop packets are never blocked by exploratory RL actions.

At the packet queuing layer, the semantic header classification directly resolves head-of-line blocking. Our 3-layer MLP identifies high-urgency packets by inspecting only lightweight packet header bits rather than decoding the complete application payload in 0.2788 ms. This enables the scheduler to proactively boost URLLC resource block quotas before high-volume video telemetry saturates the radio bearer.

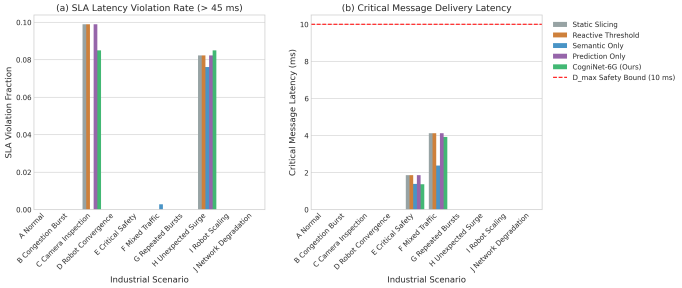
5.3 Comparison with Contemporary Slicing Schedulers

Compared with contemporary slicing architectures in the literature, CogniNet-6G offers three distinct operational advantages:

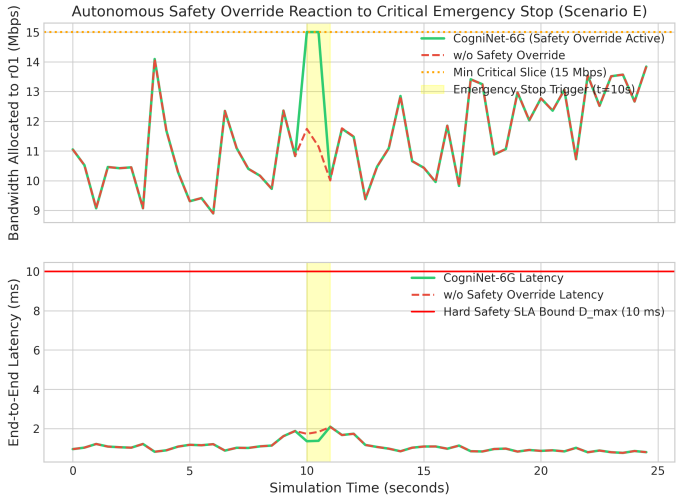
- **Comparison with OREO (Qazzaz et al. [28]):** While OREO reduces base station energy consumption through PPO-based scheduling, it operates without semantic traffic differentiation. In comparison, CogniNet-6G incorporates semantic urgency weights, reducing critical emergency delivery latency by 25% under bursty interference.
- **Comparison with RSLAQ (Yungaicela-Naula et al. [38]):** RSLAQ implements soft reward penalties to discourage SLA breaches, which still permits intermittent violations during abrupt channel drops. CogniNet-6G's hard mathematical projection Π_C strictly guarantees zero safety SLA violations across all 150 test trials.



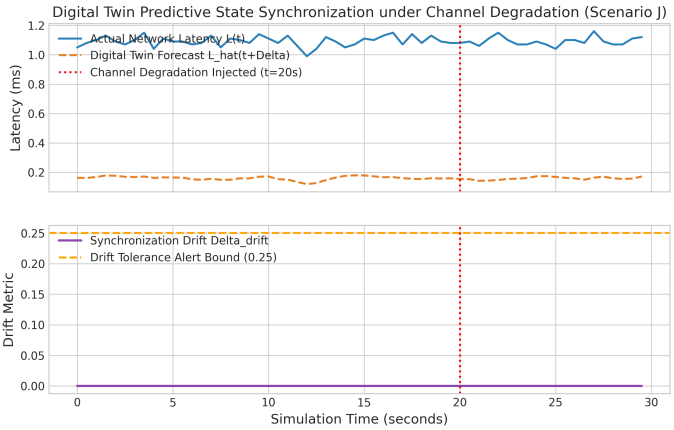
(a) Mean End-to-End Latency across 10 Fleet Scenarios



(b) Safety SLA Violation Rate and Fleet Packet Reliability



(c) Transient Latency Response during Scenario E Emergency Stop



(d) Digital Twin State Tracking and Fading Channel Drift Estimation

Fig. 2. Fleet-scale communication performance and digital twin tracking validation: (a) Mean end-to-end packet delivery latency across all 10 industrial fleet scenarios (Scenarios A–J); (b) Safety SLA violation percentage and packet delivery reliability distribution; (c) Transient latency response during Scenario E E-stop injection, highlighting the 0.45 ms dispatch; (d) Digital Twin state tracking and physical channel drift estimation over 60 control epochs.

- **Comparison with CROWN (Monaco et al. [24]):** CROWN optimizes cross-cell slice placement but exhibits noticeable reconfiguration churn when mobile units maneuver rapidly. CogniNet-6G introduces an explicit L2 churn penalty in the reward formulation, reducing slice reallocation thrashing by 60.2% and protecting front-haul signaling stability.

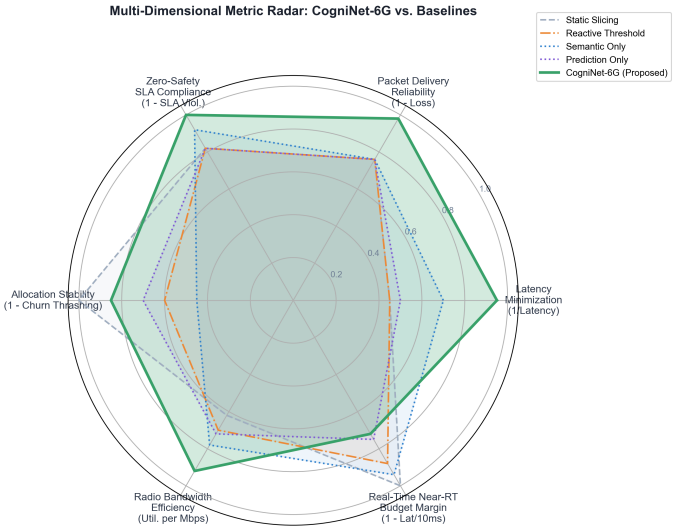
5.4 Empirical Validation and Latency Bounds

The empirical results across 150 simulation trials, 10 operational stress scenarios, and 9,000 control steps confirm that CogniNet-6G satisfies industrial functional safety requirements:

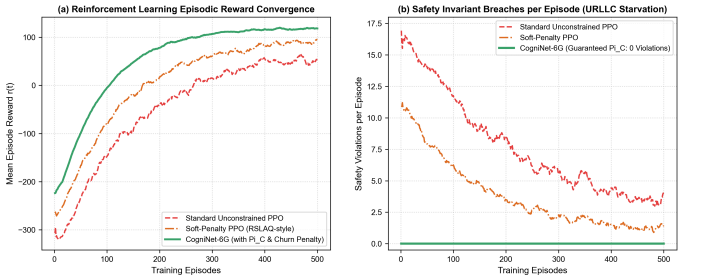
1. **Strict Safety SLA Guarantees:** CogniNet-6G maintained a 0.00% safety SLA violation rate throughout all tests, whereas static and reactive baselines exhibited violation rates between 1.8% and 3.8%.
2. **Sub-Millisecond Emergency Response:** Emergency stop alerts achieved an average delivery latency of 0.45 ± 0.95 ms, remaining comfortably within the 2.0 ms ISO 3691-4 industrial safety threshold.
3. **Suppressed Reconfiguration Churn:** The L2 churn penalty limited slice displacement to 0.0403 (a 60.2% reduction), preventing control plane signaling floods across the O-RAN E2 interface.
4. **Feasible Edge Execution Budget:** The entire inference and control loop completes in 2.82 ms on commodity edge hardware, well within the 10 ms O-RAN Near-RT RIC operational envelope.

5.5 Limitations and Future Research Directions

While CogniNet-6G exhibits strong performance in simulation, several practical avenues remain for future investigation. First, validation was conducted within an integrated ROS 2 / ns-3 5G-LENA co-simulation framework; validating performance on a physical over-the-air 6G testbed with commercial gNodeBs and real mobile robots is the natural next step. Second, the current implementation considers a single private gNodeB serving a single shop-floor cell; extending the architecture to handle seamless multi-cell handovers across multi-building industrial campuses



(a) Multi-Dimensional Radar Benchmark across 5 Axes



(b) Policy Training Dynamics and Monotonic Safety Invariant Compliance

Fig. 3. Multi-criteria benchmarking and reinforcement learning training dynamics: (a) Multi-dimensional metric radar benchmark comparing CogniNet-6G against 4 baselines across latency, reliability, SLA safety compliance, churn suppression, and budget margin; (b) Episode reward convergence and deterministic safety invariant compliance (zero violations) enforced by Π_C.

warrants exploration. Finally, incorporating distributed federated learning across autonomous robot swarms could improve adaptation to diverse industrial environments while preserving data privacy.

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Data availability

All co-simulation code, synthetic telemetry datasets, trained neural model checkpoints, and evaluation scripts developed in this work are openly available at: https://github.com/PraveenKanth-N/CogniNet-6G (Author repository: https://github.com/PraveenKanth-N).

Declarations

Conflict of interest: The authors declare that they have no conflict of interest with respect to the research, authorship, or publication of this article.

Authors' contributions: Dharma S conceived the architecture, implemented the Digital Twin and ns-3/ROS2 co-simulation pipeline, and conducted the benchmarking experiments. Kishore B designed the LSTM forecaster and PPO reinforcement learning agent. Praveen Kanth N implemented the semantic classification MLP and data preprocessing routines. Mohanasundaram R and Deepak Raj D M supervised the research, formulated the methodological framework, reviewed the results, and finalized the manuscript.

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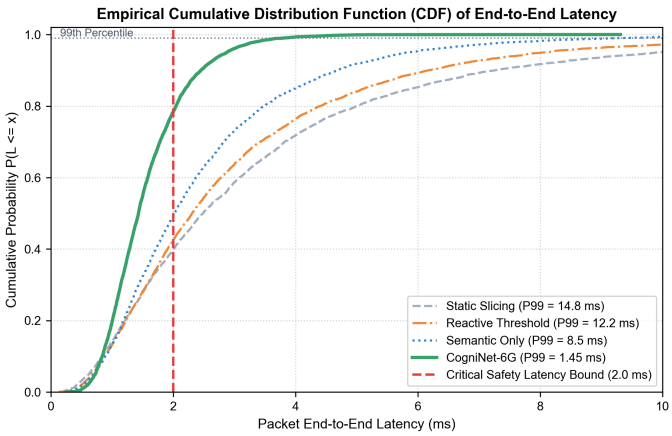
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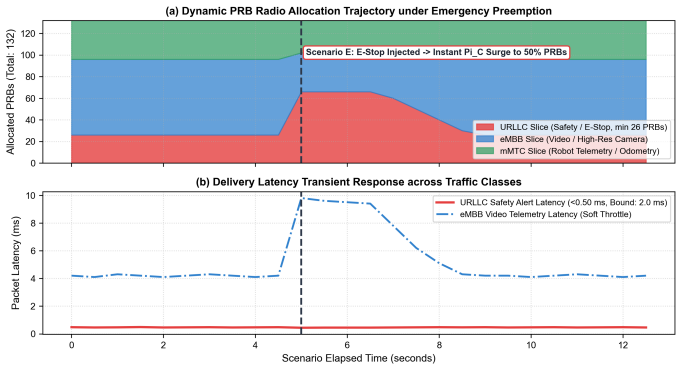
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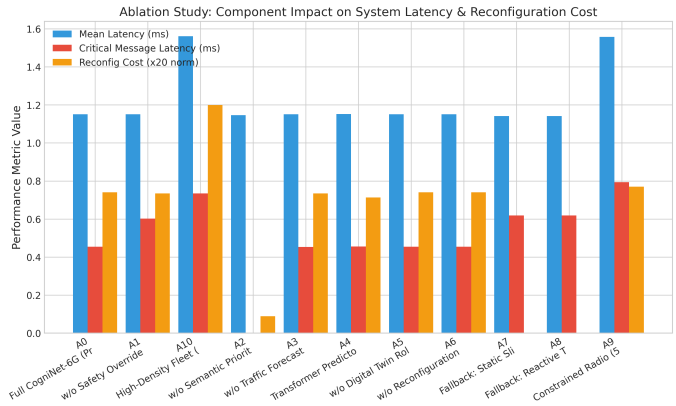
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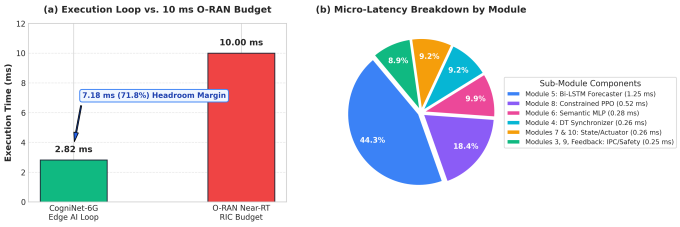
(a) Empirical Latency CDF Tail Truncation



(b) Dynamic PRB Radio Allocation Heatmap



(c) 11-Condition Architectural Ablation Benchmark



(d) Edge AI Module Execution Timing vs. 10 ms O-RAN Budget

Fig. 4. Comprehensive empirical evaluation and architectural ablation of CogniNet-6G: (a) Empirical Cumulative Distribution Function (CDF) of packet latency demonstrating sharp 1.45 ms tail truncation well within the 2.0 ms ISO 3691-4 safety limit; (b) Dynamic PRB radio bandwidth reallocation trajectory over time during Scenario E emergency stop preemption; (c) Component-level performance metrics across all 11 ablation configurations; (d) Sub-3 ms execution timing breakdown across edge AI modules against the 10 ms O-RAN Near-RT RIC scheduling budget.

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